Math 510 **Fall 2026**
Quiz 4 Solutions **Drew Armstrong**

Problem 1. Let $f : U \to V$ be a linear function. Prove that f is injective if and only if its kernel is trivial.

Recall the definition of the kernel: $ker(f) = \{\mathbf{u} \in U : f(\mathbf{u}) = \mathbf{0}\}$.

First, suppose that f is injective. To show that the kernel is trivial, suppose that $\mathbf{u} \in ker(f)$, so that $f(\mathbf{u}) = \mathbf{0}$. Since f is linear, we also have $f(\mathbf{0}) = \mathbf{0}$. Then since $f(\mathbf{u}) = f(\mathbf{0})$, and since f is injective, we conclude that $\mathbf{u} = \mathbf{0}$. We have shown that $ker(f) = \{\mathbf{0}\}$, as desired. Conversely, suppose that $ker(f) = \{\mathbf{0}\}$. To show that f is injective, suppose that $f(\mathbf{u}_1) = f(\mathbf{u}_2)$ for some $\mathbf{u}_1, \mathbf{u}_2 \in U$. Then by linearity of f we have

$$\begin{aligned} f(\mathbf{u}_1) &= f(\mathbf{u}_2) \\ f(\mathbf{u}_1) - f(\mathbf{u}_2) &= \mathbf{0} \\ f(\mathbf{u}_1 - \mathbf{u}_2) &= \mathbf{0}. \end{aligned}$$

Since the kernel is trivial, this implies that $\mathbf{u}_1 - \mathbf{u}_2 = \mathbf{0}$, and hence $\mathbf{u}_1 = \mathbf{u}_2$. We conclude that f is injective.

Problem 2. Find the unique vector $\mathbf{x}$ that minimizes the distance $\left\| \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \mathbf{x} - \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\|$.

Recall: If a matrix $A \in \mathbb{C}^{m \times n}$ has independent columns, then for any vector $\mathbf{b} \in \mathbb{C}^m$ there exists a unique vector $\mathbf{x} \in \mathbb{C}^n$ that minimizes the distance $\|A\mathbf{x} - \mathbf{b}\|$, and this vector is given by $\mathbf{x} = (A^*A)^{-1}A^*\mathbf{b}$. In our case, we have

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Since A has independent columns, the matrix A^*A is invertible. We compute

$$A^*A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

and hence

$$(A^*A)^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

Finally, we have

$$\begin{aligned} \mathbf{x} &= (A^*A)^{-1}A^*\mathbf{b} \\ &= \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\ &= \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \frac{1}{3} \begin{pmatrix} 2 \\ -1 \end{pmatrix}. \end{aligned}$$